Computer Organization
Homework 5
(Due on 2019/01/10)

Note: Please hand in the hardcopy.

1. This exercise is intended to help you understand the relationship between forwarding, hazard detection and ISA design. Problems in this exercise refer to the following sequence of instructions and assume that it is executed on a 5-stage pipelined datapath:

```
add $s1, $s2, $s1
lw $s3, 4($s1)
lw $s4, 0($s1)
sub $s3, $s4, $s3
sw $s3, 0($s1)
addi $s6, $s4, -1
sw $s6, 4($s5)
```

   (a) If there is no forwarding or hazard detection, insert nops to ensure correct execution. Please indicate the result as in the following way. (15%)

add $s1, $s2, $s1	IF	ID	EX	MEM	WB				
lw $s3, 4($s1)		IF	ID	…	…				
lw $s4, 0($s1)			IF	…					
sub $s3, $s4, $s3				…					
sw $s3, 0($s1)				…					
addi $s6, $s4, -1									
sw $s6, 4($s5)									

   (b) Is it helpful if we can rearrange the MIPS code sequence to avoid nop insertion. Note that the same execution results must be obtained. If it is possible, how to rearrange? If it is impossible, why? (10%)

   (c) For the original code sequence, what kinds of hazard detection signals should be asserted if we want to add forwarding path to solve all the hazards? (10%)

   (d) What kinds of signals are sent by the forwarding unit? Please use Fig. 4.60 or figure on P81 to explain. Note that you have to define the state of the input selection and its corresponding input port of the multiplexers. (10%)

2. Considering a direct-mapped Cache for a MIPS processor,

   (a) If the cache contains 256 8-word blocks, how many number of bits for the tag

field? (4%)

(b) What is the size of the cache memory? Please also specify the number of entries and word-length of each entry, respectively. Do remember to include the “valid” bit. (For example: 8 entries x 12 bits/entry = 96 bits.) (6%)

(c) Find the block index of the MISP instruction with address 2018 if it is in the cache. (5%)

(d) Please draw the block diagram of the cache such as the figure on page 32 of the slide. (10%)

3. Considering a 8-way set associative cache for a MIPS processor,

(a) If the cache contains 128 4-word blocks, how many number of sets in the cache? (4%)

(b) How many bits are used for the tag field? How many bits are used for the cache index? (8%)

(c) What is the size of the whole cache memory? Please also specify clearly how you derive the results. (For example: 8 entries x 12 bits/entry = 96 bits.) (8%)

(d) Please draw the block diagram of the cache such as the figure on page 45 of the slide. (10%)